

14. Implement House price Prediction using appropriate machine learning algorithm

Aim

To implement a **House Price Prediction model** using an appropriate machine learning algorithm.

Software Required

- Python 3.x
- Pandas
- Scikit-learn

Procedure

1. Create a dataset containing house area, number of bedrooms and age.
2. Define the house price as the target variable.
3. Split the data.
4. Train a Linear Regression model.
5. Predict house prices.
6. Evaluate the model using Mean Squared Error.

Code

```
import pandas as pd

from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

data = {
    'Area': [800, 1000, 1200, 1500, 1800,
            2000, 2200, 2500, 2800, 3000],
    'Bedrooms': [2, 2, 3, 3, 3, 4, 4, 4, 5, 5],
    'Age': [20, 18, 15, 12, 10, 8, 6, 5, 3, 2],
    'Price': [40, 50, 60, 75, 90,
            105, 120, 135, 150, 165]
}

df = pd.DataFrame(data)
```

```

X = df[['Area', 'Bedrooms', 'Age']]
y = df['Price']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted House Prices:")
print(y_pred)
print("\nMean Squared Error:",
      mean_squared_error(y_test, y_pred))
sample = [[2000, 4, 5]]
print("\nPredicted price:",
      model.predict(sample)[0], "Lakhs")

```

Output:

```

Predicted House Prices:
[154.46428571  48.39285714]

```

```

Mean Squared Error: 11.256377551020174

```

```

Predicted price: 101.96428571428572 Lakhs
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:100:
  warnings.warn(

```

Result

Thus, the **House Price Prediction model** was successfully implemented using Linear Regression.